SL2DT-05: Monitor & Alert Conversion

> **Series:** SL2DT — Sumo Logic to Dynatrace | **Notebook:** 5 of 11 | **Created:** April 2026 | **Last Updated:** 10/14/2026

Overview

Goal of this step: rebuild Sumo Monitors in Dynatrace, choosing the right target for each (a custom alert, a Workflow with a DQL threshold, or — on the classic surface — a metric event). This is the notebook where the biggest fidelity-vs-better-approach decisions happen.

The anti-pattern to avoid: 1:1 static-threshold lift-and-shift. It produces noisy alerts, misses real anomalies, and under-uses Dynatrace's capabilities. The pattern to prefer: custom alerts with an adaptive or seasonal model for metric-backed alerts, Workflows for complex multi-condition logic, and a static-threshold custom alert for thresholds that genuinely are static.

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Prerequisites

Requirement	Details
Audience	Core migration engineers + app-team reviewers
Inputs	`inventory/monitors.json`, translations from SL2DT-04
Dynatrace access	Platform token or OAuth client with `settings:objects:write` (custom alerts are Settings objects), `automation:workflows:write`, and `iam:service-users:use` (custom alerts that run as a service user) or `davis:analyzers:execute`; plus Grail read access to the data each alert queries (for example `storage:buckets:read`, `storage:metrics:read`, `storage:logs:read`)
Prior reading	SL2DT-04 for query translations; OPLOGS-09 for log-based alert patterns

1. What You'll Produce

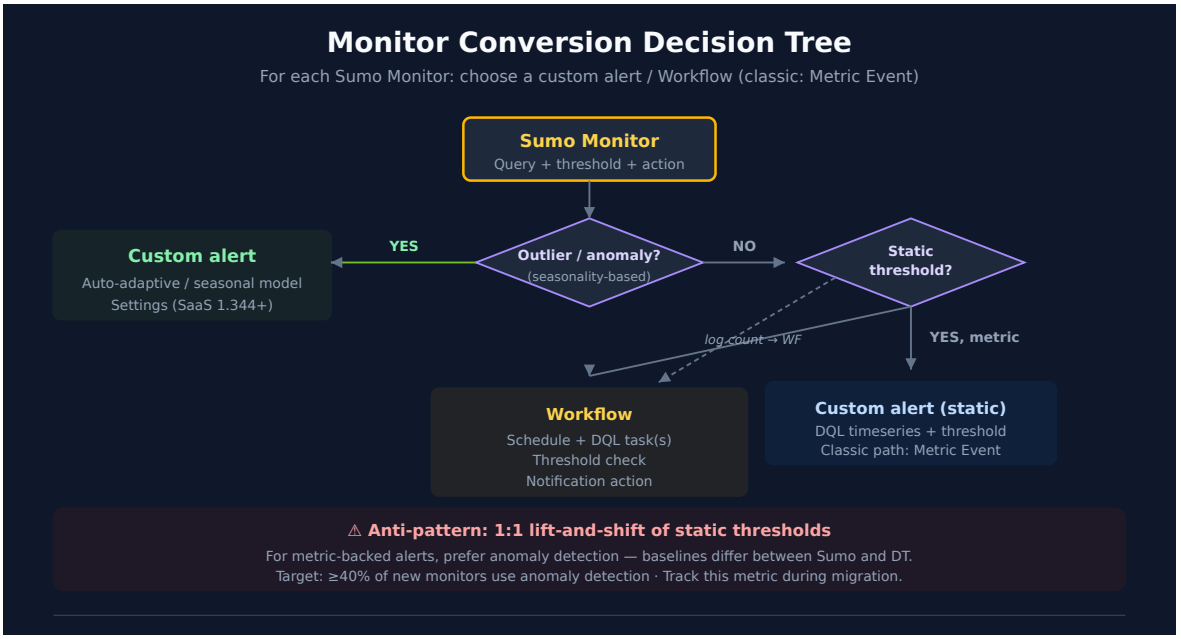
Artifact	Purpose
`monitors/decision-matrix.csv`	Per-monitor: custom alert vs Workflow vs Metric Event decision
`monitors/configs/`	Terraform + JSON for custom alerts (anomaly detectors), Workflows, Metric Events
`notifications/action-map.md`	Sumo webhook/email → DT notification target map
`monitor-rebuild-report.md`	Progress per team

2. The Monitor Conversion Decision Framework

For each Sumo monitor, pick one of three targets:

Decision Rules

Sumo Monitor Type	Target	Why
Outlier / anomaly / seasonality-based	**Custom alert** — auto-adaptive or seasonal-baseline model	Learns the baseline instead of a hand-picked number
Static threshold on metric (CPU > 90%)	**Custom alert** — static-threshold model (classic: **Metric Event**)	A DQL `timeseries` query and a threshold, as a Settings object
Static threshold on log count	**Workflow with DQL**	Query runs on schedule, threshold in task condition
Missing data alert	**Workflow**	Use `isNull` / `count == 0` checks
Multi-condition (X AND Y but not Z)	**Workflow**	Multiple DQL tasks + branching
SLO-backed alert	**SLO burn-rate alert** (if Dynatrace SLO)	Native SLO product



3. Anomaly Detection — When & How

A **custom alert** evaluates a DQL `timeseries` query with one of three models — static threshold, auto-adaptive, or seasonal baseline — and raises an event when the signal

violates it. The auto-adaptive and seasonal models learn the baseline. Best for:

- Any Sumo outlier monitor (`outlier _count window=N threshold=K`)
- Static thresholds where the "normal" value drifts seasonally (weekly cycle, business hours)
- Static thresholds that fire too often in Sumo (false positives from workload swings)

Where custom alerts live (SaaS 1.344)

From SaaS 1.344 custom alerts are created and managed in **Settings**, and the Anomaly Detection app that used to hold them is deprecated. SaaS versions reach tenants in a staged rollout: on a tenant that is not yet on 1.344, the same custom alerts are still configured in the Anomaly Detection app. Programmatically, a custom alert is a Settings object of schema `builtin:davis.anomaly-detectors` .

Configuration

```
{
  "schemaId": "builtin:davis.anomaly-detectors",
  "scope": "environment",
  "value": {
    "enabled": true,
    "title": "API error count anomaly",
    "description": "Migrated from the Sumo outlier monitor on prod/api",
    "source": "sumo-migration",
    "executionSettings": {"actor": null, "queryOffset": null},
    "analyzer": {
      "name":
"dt.statistics.ui.anomaly_detection.AutoAdaptiveAnomalyDetectionAnalyzer",
      "input": [
        {"key": "query", "value": "timeseries sum(log.sumo.api_error_count),
interval:1m, by:{dt.entity.host}"},
        {"key": "numberOfSignalFluctuations", "value": "1"},
        {"key": "alertCondition", "value": "ABOVE"},
        {"key": "alertOnMissingData", "value": "false"},
        {"key": "violatingSamples", "value": "3"},
        {"key": "slidingWindow", "value": "5"},
        {"key": "dealertingSamples", "value": "5"}
      ]
    },
    "eventTemplate": {
      "properties": [
        {"key": "event.type", "value": "CUSTOM_ALERT"},
        {"key": "event.name", "value": "API error count above baseline on
{dims:dt.entity.host.name}"},
        {"key": "event.description", "value": "Error-log count for prod/api is
above its learned baseline."},
        {"key": "dt.source_entity", "value": "{dims:dt.entity.host}"}
      ]
    }
  }
}
```

The query must return a timeseries (`timeseries` or `makeTimeseries`), set `interval:1m` , and carry no `from:` / `to:` . For a fixed threshold, use `dt.statistics.ui.anomaly_detection.StaticThresholdAnomalyDetectionAnalyzer` with a `threshold` input; for a weekly or daily pattern, `dt.statistics.ui.anomaly_detection.SeasonalBaselineAnomalyDetectionAnalyzer` with `tolerance` . `source` is free text — a shared value such as `sumo-migration` lets you filter every migrated alert in the UI.

> **Sources:**
>
> - Anomaly Detection app (DT docs) — *"Starting with Dynatrace version 1.344, custom alerts have moved to"* Settings; *"Because Anomaly Detection is deprecated, we highly recommend that you use"* Settings.
> - Set up anomaly detectors via API (DT docs) — schema `builtin:davis.anomaly-detectors`; *"The query must have a time interval explicitly set to interval:1m. You can't set timeframes using from: and to: DQL operators with this configuration."*

Sumo outlier → custom alert example

Sumo:

```
_sourceCategory=prod/api | timeslice 1m | count | outlier _count window=10 threshold=3
```

Dynatrace:
1. Create a counter metric from the matching logs in OpenPipeline (next subsection).
2. Create a custom alert on that metric with the auto-adaptive model (the JSON above).
3. Route the resulting problem to the Sumo monitor's notification target with a problem-triggered Workflow (§6).

Metric Extraction from Logs

If the Sumo monitor was log-count-based, extract a metric first. In **OpenPipeline → Logs**, open the pipeline that processes these records, select the **Metric extraction** tab, and add a **Counter metric** processor:

Field	Value
Matching condition	`sumo.source_category == "prod/api" and status == "ERROR"`
Metric key	`log.sumo.api_error_count`
Dimensions	`dt.entity.host`

Log metric keys carry the `log.` prefix. A new log metric has data only for logs ingested after it is created, so give it time to fill before the custom alert's baseline means anything.

> **Sources:** Log metrics (DT docs) — *"All metrics based on a log monitoring query have a metric key with the `log.` prefix."*, *"A created log metric is available only when new log data is ingested."*; Metric extraction (DT docs).

Classic path — metric events

Tenants that already manage alerting as classic metric events (`builtin:anomaly-detection.metric-events`) can keep that surface; it evaluates Metrics Classic keys, not DQL. An adaptive or seasonal model there needs a `METRIC_SELECTOR` query definition — *"Metric key based query definitions only support static thresholds"* — and every model needs its full set of sample counts plus an event template. New monitors in this migration should use custom alerts (above).

```
{
  "schemaId": "builtin:anomaly-detection.metric-events",
  "scope": "environment",
  "value": {
    "enabled": true,
    "summary": "API error rate anomaly",
    "queryDefinition": {
      "type": "METRIC_SELECTOR",
      "metricSelector":
"custom.metric.api_error_rate:splitBy(\"dt.entity.host\"):avg"
    },
    "modelProperties": {
      "type": "AUTO_ADAPTIVE_THRESHOLD",
      "signalFluctuation": 1.0,
      "tolerance": 4.0,
      "alertCondition": "ABOVE",
      "alertOnNoData": false,
      "samples": 5,
      "violatingSamples": 3,
      "dealertingSamples": 5
    },
    "eventTemplate": {
      "title": "API error rate anomaly",
      "description": "The API error rate is above its learned baseline.",
      "eventType": "ERROR",
      "davisMerge": true,
      "metadata": []
    }
  }
}
```

> **Sources:** builtin:anomaly-detection.metric-events schema (DT docs); Metric events (DT docs) — *"We encourage you to try Anomaly Detection"*.

Validation

```
// Confirm the log metric is producing data (it fills only from logs ingested
after it was created)
timeseries c = sum(log.sumo.api_error_count), from:-1h, by:{dt.entity.host}
| fieldsAdd c_max = arrayMax(c)
| filter c_max &gt; 0
```

4. Metric Events — Simple Static Thresholds

When the threshold truly is static and the underlying metric is well-defined:

- CPU > 90%
- Memory available < 5%
- Disk full
- Certificate expiry < 30 days

A custom alert with the static-threshold model (§3) covers this case on a DQL query and is the current path. Where you keep classic metric events, use `builtin:anomaly-detection.metric-events`:

```
{
  "schemaId": "builtin:anomaly-detection.metric-events",
  "scope": "environment",
  "value": {
    "enabled": true,
    "summary": "Host CPU &gt; 90%",
    "queryDefinition": {
      "type": "METRIC_KEY",
      "metricKey": "builtin:host.cpu.usage",
      "aggregation": "AVG",
      "entityFilter": {
        "dimensionKey": "dt.entity.host",
        "conditions": [
          {"type": "HOST_GROUP_NAME", "operator": "EQUALS", "value": "prod-
web"}
        ]
      }
    },
    "modelProperties": {
      "type": "STATIC_THRESHOLD",
      "threshold": 90.0,
      "alertCondition": "ABOVE",
      "alertOnNoData": false,
      "samples": 5,
      "violatingSamples": 3,
      "dealertingSamples": 5
    },
    "eventTemplate": {
      "title": "Host CPU &gt; 90%",
      "description": "CPU usage has been above 90% for 3 of the last 5
minutes.",
      "eventType": "RESOURCE",
      "davisMerge": true,
      "metadata": []
    }
  }
}
```

The schema's only scope is `environment`; narrow the alert to particular hosts with the `entityFilter` (here a host group), not with an entity ID as the scope. Metric events read Metrics Classic, so the key is `builtin:host.cpu.usage` — the Grail key `dt.host.cpu.usage` is for DQL.

When NOT to use Metric Events

- Threshold depends on workload (use a custom alert with an auto-adaptive or seasonal model)
- Condition involves multiple metrics (use Workflow)
- Alert requires custom notification payload (use Workflow)

5. Workflows — Complex or Multi-Condition Alerts

Workflows are the equivalent of Sumo's more complex monitors. Use when:

- Alert logic needs multiple DQL queries
- Conditional branches required
- Custom notification payload (ServiceNow incident with specific fields, etc.)
- Scheduled evaluation (every N minutes)

Workflow Structure

The schedule (cron `*/5 * * * *`, every 5 minutes) is the workflow's trigger, set in the editor rather than written into the task list. The HTTP task authenticates with a Credential Vault credential selected in its **Authentication** field.

```
tasks:
  - name: check_error_rate
    action: dynatrace.automations:execute-dql-query
    input:
      query: |
        fetch logs, from:-5m
        | filter sumo.source_category == "prod/api"
        | summarize {total = count(), errors = countIf(contains(content,
"error", caseSensitive:false))}
        | fieldsAdd error_pct = 100.0 * toDouble(errors) / toDouble(total)
  - name: check_threshold
    action: dynatrace.automations:http-function   # ServiceNow Table API — see
§6
    predecessors: [check_error_rate]
    conditions:
      states:
        check_error_rate: SUCCESS
      custom: '{{ result("check_error_rate")["records"] | length &gt; 0 and
result("check_error_rate")["records"][0]["error_pct"] &gt; 5 }}'
    input:
      method: POST
      url: https://&lt;instance&gt;.service-now.com/api/now/table/incident
      headers:
        Content-Type: application/json
      payload: |
        {
          "short_description": "API error rate {{
result('check_error_rate').records[0].error_pct }}%",
          "assignment_group": "Payments Platform",
          "priority": "2"
        }
```

A DQL task in state *Success* can still return no records, and reading `records[0]` from an empty result makes the condition error. Guard with `| length > 0`, as above.

> **Sources:** DQL query action (DT docs) — *"Accessing a specific record without first checking the array length will cause a condition error."*

Scheduled Search → Workflow

A Sumo scheduled search (query + schedule + action) maps directly:

Sumo Scheduled Search Field	Workflow Equivalent
Query	**Execute DQL Query** task (`dynatrace.automations:execute-dql-query`)
Schedule (run every N)	Schedule trigger with a cron rule, set in the editor
Alert condition	`conditions.custom` on the notification task (a Jinja expression; check `records | length > 0` before reading `records[0]`)
Webhook action	HTTP Request task (`dynatrace.automations:http-function`) or a connector action
Email action	**Send email** task (`dynatrace.email:send-email`)

DQL in Workflow Tasks

The translated DQL from SL2DT-04 goes directly here. Wrap long queries in `|` to preserve formatting:

```
tasks:
  - name: find_slow_transactions
    action: dynatrace.automations:execute-dql-query
    input:
      query: |
        fetch logs, from:-5m
        | filter sumo.source_category == "prod/api"
        | parse content, "LD? 'latency=' INT:latency"
        | filter latency &gt; 2000
        | summarize c = count(), by:{http.path}
        | sort c desc
        | limit 20
```

6. Rebuilding Notification Actions

Every Sumo monitor has one or more actions. Map each to a Dynatrace notification target.

Sumo Action	Dynatrace Target	Notes
Email	**Send email** task (`dynatrace.email:send-email`)	Same recipient list; at most ten per field
Webhook → ServiceNow	HTTP Request task to the Table API, or ServiceNow **Create Incident** (`dynatrace.servicenow:snow-create-incident`)	Rebuild incident payload
Webhook → Slack	Slack **Send message** task (`dynatrace.slack:slack-send-message`)	Rebuild message format
Webhook → PagerDuty	PagerDuty **Send event** task (`dynatrace.pagerduty:send-event`)	Rebuild event payload
Webhook → custom HTTP	HTTP Request task (`dynatrace.automations:http-function`)	
Mobile push	Workflow + Dynatrace Mobile app	

ServiceNow Integration — Specific Patterns

ServiceNow is the most common target and the highest-risk translation (wrong field mapping → misrouted incidents).

Sumo webhook payload:

```
{
  "incident": {
    "short_description": "{{Name}}: {{Description}}",
    "u_affected_service": "{{ClusterName}}",
    "assignment_group": "{{Team}}",
    "priority": "{{Priority}}"
  }
}
```

Dynatrace Workflow equivalent:

```
- name: create_servicenow_incident
  action: dynatrace.automations:http-function
  input:
    method: POST
    url: https://&lt;instance&gt;.service-now.com/api/now/table/incident
    # Authentication: select a Credential Vault credential (Basic) in the
task's
    # Authentication field. Never put the credential in a static Authorization
header.
    headers:
      Content-Type: application/json
    payload: |
      {
        "short_description": "API error rate {{
result('check_error_rate').records[0].error_pct }}%",
        "u_affected_service": "{{ result('check_error_rate').records[0]
['dt.entity.service'] }}",
        "assignment_group": "Payments Platform",
        "priority": "2"
      }
```

The request body goes in `payload`, and dotted field names need bracket access (`records[0]['dt.entity.service']`). The HTTP route keeps custom fields such as `u_affected_service`. The ServiceNow Connector's **Create Incident** action (`dynatrace.servicenow:snow-create-incident`) is the alternative when the standard fields are enough. It takes named inputs (`shortDescription`, `description`, `impact`, `urgency`, `category`, `group`, `correlationId`, …) rather than a raw body. For the query above, `u_affected_service` stays empty unless the query groups by `dt.entity.service`.

> **Sources:** HTTP request action (DT docs) — *"Payload : The payload of the HTTP request."*, *"We strictly advise against providing any static Authorization header and therefore, leak a secret."*; wftpl_sample_servicenow_incident_man.yaml (Dynatrace GitHub) — *"action: dynatrace.servicenow:snow-create-incident"*.

Verify with a test incident before flipping production traffic.

7. Handling Rare Alert Classes

Missing Data Alerts

Sumo:

```
_sourceCategory=prod/heartbeat | count
| where _count == 0
```

Dynatrace Workflow:

```
tasks:
  - name: check_heartbeat
    action: dynatrace.automations:execute-dql-query
    input:
      query: |
        fetch logs, from:-10m
        | filter sumo.source_category == "prod/heartbeat"
        | summarize c = count()
  - name: alert_if_missing
    action: dynatrace.email:send-email
    predecessors: [check_heartbeat]
    conditions:
      custom: '{{ result("check_heartbeat")["records"] | length == 0 or
result("check_heartbeat")["records"][0]["c"] == 0 }}'
    input:
      to: ["oncall@example.com"]
      cc: []
      bcc: []
      subject: "No heartbeat from prod/heartbeat"
      content: "No heartbeat logs from prod/heartbeat in the last 10 minutes."
```

An ungrouped `summarize c = count()` returns one record with `c = 0` when nothing matches, so the second clause is the one that fires; the `length == 0` clause treats an empty result as missing data too, instead of erroring.

Change Detection Alerts

Sumo:

```
_sourceCategory=audit | logcompare timeshift=24h
```

Dynatrace: Change detection or two-fetch comparison workflow. See OPLOGS-08.

Multi-Metric Correlation

Sumo: two separate monitors with same target; Dynatrace: one Workflow with two DQL tasks + AND condition.

8. Tuning & Noise Reduction

Expect more alerts in the first week post-cutover — baselines aren't learned yet, and Workflow thresholds may be too aggressive.

First-week discipline

1. Monitor alert volume daily. Dashboard the total count of detected problems + Workflow triggers.
2. For every alert, triage: real issue, false positive, or threshold too tight?
3. Adjust thresholds daily until noise drops below Sumo baseline.

anomaly detection — let it learn

Dynatrace Intelligence needs ~2 weeks of data to build a stable baseline. During week 1–2, expect more sensitivity alerts. Keep them in a separate "tuning" severity class if possible; don't page on-call for them.

Checking alert volume

```
// Alert volume over time — compare Sumo baseline
// dt.davis.problems holds one record per problem; `fetch events | filter
event.kind == "DAVIS_PROBLEM"`
// returns every update of every problem and over-counts many times over.
fetch dt.davis.problems, from:-7d
| makeTimeseries problems = count(), interval:24h, time:event.start
```

Silence during load tests

Configure a maintenance window before load tests, CI/CD deployments, or controlled outages. In Latest Dynatrace the Settings schema is `builtin:maintenance-windows`, and a window is scoped with DQL matcher expressions on Grail primary fields and tags. Latest Dynatrace always detects problems during a window — there is no option to disable detection — so muting is done in the notification path: problems raised during a window carry `maintenance.is_under_maintenance = true`, and a problem-triggered Workflow skips them when its trigger's additional custom filter query is `maintenance.is_under_maintenance != true`.

On a classic tenant, maintenance windows use the classic schema `builtin:alerting.maintenance-window`, whose `suppression` option chooses between *Detect problems and alert*, *Detect problems but don't alert*, and *Disable problem detection during maintenance*. Classic windows are not migrated automatically when you upgrade; recreate them as Latest Dynatrace windows.

A scheduled DQL workflow (§5, §7) is not a problem trigger, and the maintenance-window pages do not say it consults the window. In community practice, disable those workflows for the window, or move the alert to a custom alert plus a problem-triggered workflow so the filter above applies.

> **Sources:** Upgrade from classic maintenance windows (DT docs) — *"Latest Dynatrace always detects problems."*, *"There is no equivalent to disabling problem detection entirely."*, *"Window scope uses DQL matcher filter criteria on Grail primary fields and tags."*, *"Classic maintenance windows are not migrated automatically"*; builtin:maintenance-windows schema (DT docs); builtin:alerting.maintenance-window schema (DT docs); Use maintenance window data in Grail (DT docs) — *"maintenance.is_under_maintenance != true"*.

Severity mapping

Dynatrace has no P1/P2/P3 severity. A problem carries an `event.category` (for example `AVAILABILITY`, `ERROR`, `SLOWDOWN`, `RESOURCE_CONTENTION`, `CUSTOM_ALERT`), and routing happens in the problem-triggered Workflow. Carry the Sumo priority across as an extra key-value property on the custom alert's event template (for example `sumo.priority`), confirm it appears on the resulting problem in your tenant, and route on it; FAQ-21 covers the routing pattern.

Sumo Priority	Route on	Notification
Critical	`sumo.priority == "critical"`, or `event.category` `AVAILABILITY` / `ERROR`	PagerDuty + ServiceNow
High	`sumo.priority == "high"`	ServiceNow
Warning	`sumo.priority == "warning"`	Email
Info	(no alert)	Logged only

9. Step Exit Criteria

G5 — Monitors Rebuilt

- [ ] Every Sumo monitor has a decision logged (custom alert / Workflow / Metric Event / retire)
- [ ] ≥40% of new monitors use anomaly detection (not static thresholds) — track this metric
- [ ] Top-10 business-critical monitors validated end-to-end (fires on known-bad condition, doesn't fire on normal)
- [ ] Notification integrations (ServiceNow, Slack, PagerDuty) tested with live test incidents
- [ ] First-week alert volume ≤ Sumo baseline
- [ ] App teams trained on monitor rebuild (train-the-trainer — see SL2DT-01)

Next step: SL2DT-06 — Dashboard Conversion (rebuild dashboards using the translation output).

---

10. References

Dynatrace alerting and intelligence

- Anomaly detection (DT docs)
- Workflows (DT docs)

- Notifications and alerting (DT docs)
- Problems app (DT docs)
- Maintenance windows (DT docs)

Sumo Logic monitor and alert reference (source)

- Sumo Logic monitors (Sumo Logic docs)

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